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教育经历

2023.09 – 至今 **清华大学**, 自动化系, 博士研究生
研究方向：基础模型架构、Test-Time Training、序列建模、持续学习

2019.09 – 2023.07 **清华大学**, 自动化系, 本科
GPA：3.98/4.00；专业排名：2/176

2019 **高考陕西省第6名**

学术论文

CVPR 2026 Oral
Best Paper Finalist
15/16092

ViT^3: Unlocking Test-Time Training in Vision
Dongchen Han, Yining Li, Tianyu Li, Zixuan Cao, Ziming Wang, Jun Song, Yu Cheng, Bo Zheng, Gao Huang
- 在视觉领域系统性开启测试时训练范式（Test-Time Training, TTT），提出六条设计原则
- 提出视觉TTT模型Vision Test-Time Training（ViT^3），以线性复杂度精确建模长序列

AAAI 2026

Vision Transformers are Circulant Attention Learners
Dongchen Han*, Tianyu Li*, Ziyi Wang, Gao Huang (*Equal Contribution)
- 发现视觉Transformer中注意力分布时常表现为BCCB（Block Circulant Matrices with Circulant Blocks, BCCB）特殊矩阵模式
- 利用此性质提出循环注意力，能够以对数线性复杂度实现全局自注意力计算

NeurIPS 2024

Demystify Mamba in Vision: A Linear Attention Perspective
Dongchen Han, Ziyi Wang, Zhuofan Xia, Yizeng Han, Yifan Pu, Chunjiang Ge, Gao Huang
- 建立Mamba与线性注意力的统一数学表述框架，揭示二者内在的理论联系
- 发现Mamba的核心优势来自遗忘门与宏观结构，并融合两类范式的优势，提出先进线性注意力机制MILA

NeurIPS 2024

Bridging the Divide: Reconsidering Softmax and Linear Attention
Dongchen Han*, Yifan Pu*, Zhuofan Xia*, Yizeng Han, Xuran Pan, Xiu Li, Jiwen Lu, Shiji Song, Gao Huang (*Equal Contribution)
- 发现线性注意力存在非单射问题，不同Query可能得到相同的注意力分数
- 提出单射线性注意力，用减法归一化替代除法归一化，简单有效

ECCV 2024

Agent Attention: On the Integration of Softmax and Linear Attention
Dongchen Han*, Tianzhu Ye*, Yizeng Han, Zhuofan Xia, Shiji Song, Gao Huang (*Equal Contribution)
- 在注意力计算中引入一组额外的代理向量，实现线性复杂度自注意力计算
- 无需训练即可直接加速Stable Diffusion 1.8倍，同时提升生成质量

ICCV 2023

FLatten Transformer: Vision Transformer using Focused Linear Attention
Dongchen Han*, Xuran Pan*, Yizeng Han, Shiji Song, Gao Huang (*Equal Contribution)
- 解决线性注意力聚焦能力不足和特征多样性缺失问题
- 提出聚焦函数，不引入任何额外参数，有效提升线性注意力的聚焦能力；提出采用深度卷积提升等效注意力矩阵秩的方案，简单有效。相关设计被诸多后续工作沿用

ICML 2026 **SiameseNorm: Breaking the Barrier to Reconciling Pre/Post-Norm**
Tianyu Li*, **Dongchen Han***, Zixuan Cao, Haofeng Huang, Mengyu Zhou, Ming Chen, Erchao Zhao, xiaoxi jiang, Guanjun Jiang, Gao Huang (*Equal Contribution)
- 提出高稳定性、高表征能力的Norm方案，能够直接替换Pre-和Post-Norm
- 在语言模型上优势显著，验证规模达15A2B MoE模型

ICML 2026 **Linearizing Vision Transformer with Test-Time Training**
Yining Li*, **Dongchen Han***, Zeyu Liu*, Hanyi Wang, Yulin Wang, Gao Huang (*Equal Contribution)
- 提出基于测试时训练的预训练Transformer注意力计算线性化方法
- 仅4卡H20 1小时微调，将预训练Stable Diffusion注意力线性化，性能持平、生图速度提升1.5倍

CVPR 2024 **GSVA: Generalized Segmentation via Multimodal Large Language Models**
Zhuofan Xia*, **Dongchen Han***, Yizeng Han, Xuran Pan, Shiji Song, Gao Huang (*Equal Contribution)
- 解决广义指代分割中的多目标与空目标问题
- 引入拒绝分割token [rej]，以处理不存在目标物体的情况；多次复用相同的分割token [seg]，以处理多个目标物体的情况。方法简单有效，模型性能在发布时处于SOTA

ICCV 2023 **Dynamic Perceiver for Efficient Visual Recognition**
Yizeng Han*, **Dongchen Han***, Zeyu Liu, Yulin Wang, Xuran Pan, Shiji Song, Gao Huang (*Equal Contribution)
- 缓解多出口神经网络中的特征早熟问题
- 设计双路径架构，将特征建模与分类头解耦，使早期出口能够进一步提升后续出口的性能

实习经历

2026.01 – 至今 **字节跳动Seed**, *Top Seed Intern*, 北京
大语言模型架构研究，重点关注上下文扩展与持续学习

2024.05 – 2025.12 **阿里巴巴未来生活实验室**, *Research Intern*, 北京
高效序列建模机制研究

2022.06 – 2022.08 **旷视科技**, *Research Intern*, 北京
参与NeurIPS 2022视觉域适应挑战赛

学术服务与学生工作

2024.11 – 至今 CVPR, NeurIPS, ICCV, ECCV, AAAI审稿人
2022.05 – 2023.05 清华大学自动化系学生科协主席
2022.05 – 2023.05 清华大学自动化系团委委员

荣誉与奖项

2026 CVPR 2026 Best Paper Finalist（15/16092）
2025 国家奖学金（Top 2%）
2023 清华大学优秀毕业生
2023 北京市优秀毕业生
2023 清华大学优秀本科毕业论文
2023 清华大学未来学者奖学金
2022 商汤奖学金（全国30名）
2022 清华大学优秀学生干部
2021 & 2022 蒋南翔奖学金（Top 1%，校内评价高于国奖）
2020 国家奖学金（Top 2%）
2019 清华大学新生二等奖学金（高考省前10名）

Dongchen Han

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Education

2023.09 – Present **Tsinghua University**, *PhD Candidate*, Department of Automation, Beijing, China
Research Interests:
Foundation Model, Architecture Design, Test-Time Training, Continue Learning

2019.09 – 2023.07 **Tsinghua University**, *Bachelor Degree*, Department of Automation, Beijing, China
GPA: 3.98/4.00 | Rank: 2/176 in the Department

2019 **Ranked 6th in Gaokao (Shaanxi Province)**

Research Publications

CVPR 2026 Oral
Best Paper Finalist
15/16092

ViT^3: Unlocking Test-Time Training in Vision

Dongchen Han, Yining Li, Tianyu Li, Zixuan Cao, Ziming Wang, Jun Song, Yu Cheng, Bo Zheng, Gao Huang

- Conducted the first systematic empirical study on Test-Time Training (TTT) designs for vision, establishing design principles
- Proposed Vision Test-Time Training (ViT^3), a pure TTT architecture with linear $O(N)$ complexity and parallelizable computation

AAAI 2026

Vision Transformers are Circulant Attention Learners

Dongchen Han*, Tianyu Li*, Ziyi Wang, Gao Huang (*Equal Contribution)

- Revealed the inherent phenomenon that self-attention matrices in vision Transformers naturally approximate Block Circulant matrices with Circulant Blocks (BCCB), a structured matrix enabling $O(NlogN)$ computation via 2D discrete Fourier transform
- Proposed Circulant Attention by modeling attention maps as BCCB matrices with an efficient DFT-based calculation algorithm, achieving favorable efficiency-expressiveness trade-off

NeurIPS 2024

Demystify Mamba in Vision: A Linear Attention Perspective

Dongchen Han, Ziyi Wang, Zhuofan Xia, Yizeng Han, Yifan Pu, Chunjiang Ge, Gao Huang

- Established a unified mathematical representation framework for Mamba and linear attention, revealing their intrinsic theoretical correlation
- Identified the core advantages of Mamba originate from the forget gate and macro structure, and proposed MILA: advanced linear attention via integrating strengths of both paradigms

NeurIPS 2024

Bridging the Divide: Reconsidering Softmax and Linear Attention

Dongchen Han*, Yifan Pu*, Zhuofan Xia*, Yizeng Han, Xuran Pan, Xiu Li, Jiwen Lu, Shiji Song, Gao Huang (*Equal Contribution)

- Addressed the critical non-injective problem of linear attention, where distinct queries are mapped to identical attention distributions
- Proposed injective linear attention by replacing division normalization with subtraction, achieving significant performance gain with minimal modification

ECCV 2024

Agent Attention: On the Integration of Softmax and Linear Attention

Dongchen Han*, Tianzhu Ye*, Yizeng Han, Zhuofan Xia, Shiji Song, Gao Huang (*Equal Contribution)

- Mitigated the inherent redundancy in self-attention, where similar queries generate close attention weights and redundant outputs
- Introduced a small set of agent tokens to aggregate global information first, then broadcast to all queries, effectively reducing computation complexity while preserving strong capability

ICCV 2023 **FLatten Transformer: Vision Transformer using Focused Linear Attention**
Dongchen Han*, Xuran Pan*, Yizeng Han, Shiji Song, Gao Huang (*Equal Contribution)
- Tackled the insufficient focusing ability of linear attention and feature diversity loss caused by low-rank attention matrices
- Proposed a parameter-free focusing function to enhance attention focusing capability, and a depth convolution-based scheme to boost the rank of equivalent attention matrix, widely adopted by subsequent works

ICML 2026 **SiameseNorm: Breaking the Barrier to Reconciling Pre/Post-Norm**
Tianyu Li*, **Dongchen Han***, Zixuan Cao, Haofeng Huang, Mengyu Zhou, Ming Chen, Erchao Zhao, xiaoxi jiang, Guanjun Jiang, Gao Huang (*Equal Contribution)
- Revisited the trade-off between Pre- and Post-Norm in deep Transformers
- Proposed SiameseNorm, a multi-stream normalization design which delivers consistent gains across dense/MoE language models and vision transformers with negligible overhead.

ICML 2026 **Linearizing Vision Transformer with Test-Time Training**
Yining Li*, **Dongchen Han***, Zeyu Liu*, Hanyi Wang, Yulin Wang, Gao Huang (*Equal Contribution)
- Tackled the weight inheritance barrier in linearizing pretrained Transformers
- Proposed a test-time training-based framework which linearized Stable Diffusion 3.5 with only 1-hour fine-tuning on 4 H20 GPUs. The resulting model preserved nearly 100% performance with significant efficiency gains.

CVPR 2024 **GSVA: Generalized Segmentation via Multimodal Large Language Models**
Zhuofan Xia*, **Dongchen Han***, Yizeng Han, Xuran Pan, Shiji Song, Gao Huang (*Equal Contribution)
- Resolved the multi-target and null-target challenges in generalized referring segmentation
- Introduced a rejection token [rej] for non-existent targets and a reusable segmentation token [seg] for multi-target scenarios, achieving state-of-the-art (SOTA) performance upon release

ICCV 2023 **Dynamic Perceiver for Efficient Visual Recognition**
Yizeng Han*, **Dongchen Han***, Zeyu Liu, Yulin Wang, Xuran Pan, Shiji Song, Gao Huang (*Equal Contribution)
- Alleviated the feature premature problem in multi-exit neural networks
- Designed a dual-path architecture to decouple feature modeling and classification heads, enabling early exits to boost the performance of later ones

Internship Experience

2026.01 – Present **ByteDance Seed**, *Top Seed Intern*, Beijing, China
Conducted research on LLM architecture, focusing on context scaling and continue learning

2024.05 – 2025.12 **Alibaba Future Life Lab**, *Research Intern*, Beijing, China
Conducted research on advanced sequence modeling mechanisms

2022.06 – 2022.08 **Megvii**, *Research Intern*, Beijing, China
Participated in the Visual Domain Adaptation Challenge of NeurIPS 2022

Service

2024.11 – Present **Reviewer**, *CVPR, NeurIPS, ICCV, ECCV, AAAI*

2022.05 – 2023.05 **Chairman**, *Student Association for Science and Technology of the Department of Automation*, Tsinghua University, Beijing, China

2022.05 – 2023.05 **Committee Member**, *Youth League Committee of the Department of Automation*, Tsinghua University, Beijing, China

Honors & Awards

2026 CVPR 2026 Best Paper Finalist (15/16092)

2025 National Scholarship (Top 2% of students)

2023 Outstanding Graduate of Tsinghua University

2023 Outstanding Graduate of Beijing Municipality

2023 Outstanding Undergraduate Thesis, Tsinghua University

2023 Future Scholar Scholarship, Tsinghua University

2022 SenseTime Scholarship (Awarded to only 30 outstanding students nationwide)

2022 Outstanding Student Cadre of Tsinghua University

2021 & 2022 Jiang Nanxiang Scholarship (Top 1% of students, higher honor than National Scholarship in Tsinghua University)

2020 National Scholarship (Top 2% of students)

2019 Second-Class Freshman Scholarship of Tsinghua University (Top 10 in provincial college entrance examination)